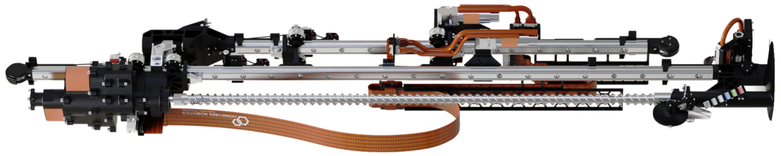
Autonomous Fault and Anomaly Detection in Extraterrestrial Drilling Using Telemetry and Acoustic Sensing

Greta Brown[1], Sarah Boelter[1], Rene Mai[2], Thomas Stucky[3], Maria Gini[1], Brian J. Glass[4]

1: University of Minnesota, Minneapolis, MN 2: Rensselaer Polytechnic Institute 3: KBR, Moffett Field, CA 4: NASA Ames Research Center, Moffett Field, CA



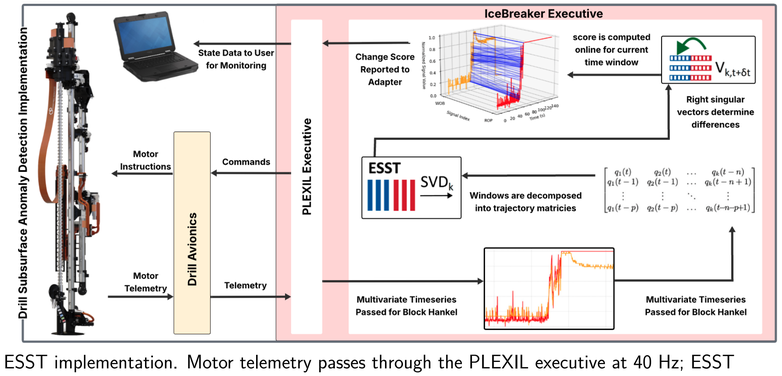
Abstract

Subsurface sampling can detect biomarkers protected from harsh surface conditions of planetary bodies and is a crucial step in the search for extraterrestrial life, but environmental, computing, energy, and operational constraints require robotic agents to complete tasks without human supervision. We validate work using online model-free time-series subspace analysis method on drill telemetry, Enhanced Singular Spectrum Transformation (ESST) [1], for noisy and sparse data to estimate drilling faults from drill avionics telemetry in Haughton Crater in the Canadian High Arctic. We also present a preliminary acoustic analysis of TRIDENT drilling, which shows that faulty drilling produces up to 100 times more high-frequency (8–15 kHz) energy than normal drilling, and outline plans to capture this signature as an additional stream or complementary metric to ESST with a piezoelectric contact microphone suited to spaceflight.

Introduction

Faults tend to raise variables like motor torque τ, weight on bit ϕ, while the rate of penetration ρ is roughly zero. The Auger Velocity ω of the drill string tends to have sudden dips from getting caught or stuck briefly with increased friction in the borehole leading up to a fault [2]. Operators also hear faults as drilling into a fault is often accompanied by audible squeaking.

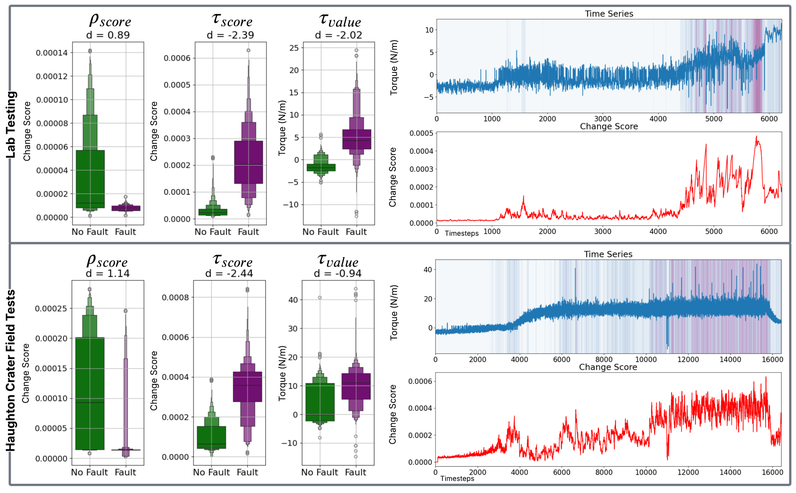
Implementation on TRIDENT Drill



ESST implementation. Motor telemetry passes through the PLEXIL executive at 40 Hz; ESST decomposes each data window into a Hankel matrix and passes the resulting change score to the user.

Results

Results reinforce prior findings [2] that ESST online telemetry monitoring improves anomaly and fault detection despite limited data. For the Haughton Crater evaluation, results from ρ score, τ score, and τ telemetry indicate effective subsurface anomaly and fault detection.



ESST results from lab (top) and Haughton Crater field (bottom) testing. Left: change-score distributions and Cohen's D for faulty vs. non-faulty data. Right: motor torque τ over a fault with its online change score; purple shading marks change-score severity.

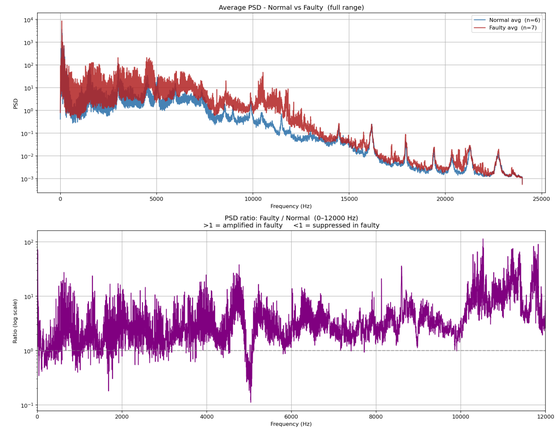
Value	Cohen's D Value	
Location	Lab	Field Site
τ value	-2.02	-0.94
τ score	-2.39	-2.44
ρ score	0.89	1.14

Value	Average F1 Score	
Location	Lab	Field Site
τ score	0.82	0.75
ρ score	0.68	0.78

Cohen's D and F1 scores for ESST results. All of our datasets had a Cohen's D score greater than ± 0.8, indicating faulty telemetry data for our datasets had a significant difference from our non-faulty telemetry data. All of our datasets had F1 scores greater than 0.6, indicating good recall.

Preliminary Acoustic Analysis

In May 2025 we recorded TRIDENT drilling with an air microphone in a lab setting and compared the average power spectral density (Welch method) of 6 normal and 7 faulty holes. Normal percussive drilling concentrates its energy in percussion harmonics below a few hundred Hz. Faulty holes are louder across most of the spectrum, and most distinctly at **8–15 kHz**, consistent with the squeaking heard during faults.

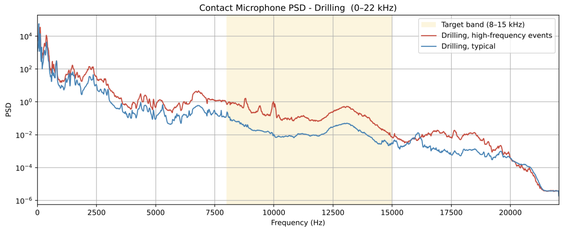


Air-microphone results. **Top:** average PSD of normal (blue) and faulty (red) holes, 0–24 kHz. Faulty drilling carries more energy up to about 15 kHz; above that the two converge. **Bottom:** ratio of faulty to normal PSD, 0–12 kHz (above 1 = amplified in faulty holes). Most of the band is amplified 1–10×, while 10–12 kHz is amplified 10–100×. Preliminary: whole-hole averages dilute brief faults.

Future Work: Contact Microphone

An air microphone picks up ambient noise and needs an atmosphere, so it cannot fly on most missions. A piezoelectric contact microphone mounted on the drill senses the fault signature as structure-borne vibration. Initial July 2026 recordings show it captures the **8–15 kHz** band. Next, we will record controlled faults (sampling at ≥ 48 kHz) and evaluate the acoustic signal in two ways: on its own, as a standalone fault detector that may call for a new method, and with ESST, either as a complementary detector or as an additional telemetry stream within ESST.





Left: piezoelectric contact microphone attached on drill stand leg. **Right:** PSD of an initial July 2026 recording; red averages the 30 one-second windows with the most energy above 8 kHz.

References

[1] S. Boelter, G. Brown, L. Weber, T. Stucky, B. Glass, and M. Gini, "Model-free subsurface anomaly detection using subspace analysis techniques for sparse telemetry for extraterrestrial drilling robots," *Proc. IEEE Int'l Conf. on Robotics and Automation (ICRA)*, 2026.

[2] S. Boelter, L. Weber, R. Lenz, B. Glass, and M. Gini, "Fault prediction in drilling using subspace analysis techniques," *Intelligent Autonomous Systems 19, Proc. 19th Int'l Conf IAS-19*, 2025.